

Taeseong Kim

INTEGRATED PH.D. STUDENT · CRYPTOGRAPHY

27-441, Gwanak-ro 1, Gwanak-gu, Seoul, South Korea

☎ | ✉ kts1023@snu.ac.kr | 🏠 taeseongkim.github.io

Overview

I am an Integrated Ph.D. student at the Department of Mathematical Sciences, Seoul National University (SNU), Republic of Korea. My advisor is Prof. Jung Hee Cheon. I work on cryptography, currently focusing on Homomorphic Encryption (HE) and its applications, including privacy-preserving machine learning.

Publications

Authors are listed in alphabetical order by last name.

CONFERENCE PUBLICATIONS

Youngjin Bae, Jung Hee Cheon, Minsik Kang, Taeseong Kim[‡]

[C03] HIGH-THROUGHPUT AES TRANSCIPHERING USING CKKS: LESS THAN 1MS

[‡] denotes corresponding author

WAHC

2025

Dongwhan Rho[†], Taeseong Kim[†], Minje Park, Jung Woo Kim, Hyunsik Chae, Jung Hee Cheon, Ernest K. Ryu

[C02] ENCRYPTION-FRIENDLY LLM ARCHITECTURE

[†] denotes co-first author

ICLR

2025

Wonhee Cho, Guillaume Hanrot, Taeseong Kim, Minje Park, Damien Stehlé

[C01] FAST AND ACCURATE HOMOMORPHIC SOFTMAX EVALUATION

ACM CCS

2024

JOURNAL PUBLICATIONS

Jung Hee Cheon, Minsik Kang, Taeseong Kim, Junyoung Jung, Yongdong Yeo

[J01] BATCH INFERENCE ON DEEP CNNs WITH FULLY HOMOMORPHIC ENCRYPTION USING CHANNEL-BY-CHANNEL CONVOLUTIONS

IEEE TDSC

Aug. 2024

MANUSCRIPTS

Jung Hee Cheon, Seungwan Hong, Minsik Kang, Jonghyun Kim, Taeseong Kim, Changmin Lee, Junho Lee

[M01] MRFHE: MIXED-RADIX FULLY HOMOMORPHIC ENCRYPTION WITH BETTER BATCH BOOTSTRAPPING

IACR ePrint 2026/853

2026

Taeseong Kim, Jonghoo Lee, Taeyeong Noh, Jung Hee Cheon, Guillaume Hanrot

[M02] LATENCY-AWARE, HIGH-THROUGHPUT HOMOMORPHIC AES EVALUATION WITH CKKS

IACR ePrint 2026/1209

2026

Talks

2025

Korea Mathematics Society Spring Meeting

HIGH-THROUGHPUT AES TRANSCIPHERING USING CKKS: LESS THAN 1MS

Daejeon, South Korea

Apr. 2025

2024

Korea Mathematics Society Fall Meeting

ENCRYPTION-FRIENDLY LLM ARCHITECTURE

Suwon, South Korea

Oct. 2024

Korea Mathematics Society Spring Meeting

FAST AND ACCURATE HOMOMORPHIC SOFTMAX EVALUATION

Daejeon, South Korea

May 2024

2023

Korea Mathematics Society Fall Meeting

BATCH INFERENCE ON DEEP CONVOLUTIONAL NEURAL NETWORKS WITH FULLY HOMOMORPHIC ENCRYPTION USING CHANNEL-BY-CHANNEL CONVOLUTIONS

Seoul, South Korea

Oct. 2023

Honors & Awards

AWARDS

- Oct. 2024 **Excellent Prize (3rd Place)**, National Cryptography Contest, NSRI (for [C01])
- Oct. 2023 **Special Prize**, National Cryptography Contest, NSRI (for [J01])
- Dec. 2019 **First Place Scholarship**, Software-Focused University Program (Undergraduate)

HONORS & SCHOLARSHIPS

- 2025–2026 **SNU Teaching and Research Support Scholarship**, Seoul National University
- 2022–2024 **BK 21+ Scholarship**, Ministry of Education of Korea
- 2018–2021 **National Scholarship for Outstanding Students in Science & Engineering**, Korea Student Aid Foundation

Experiences

TEACHING

Seoul National University

Seoul, Republic of Korea

TEACHING ASSISTANT, DEPARTMENT OF MATHEMATICAL SCIENCES

Mar. 2022 - Present

- Introduction to Cryptography
- Calculus for Humanities and Social Sciences 1
- Calculus Practice 2
- Calculus Practice 1
- Differential Equation and Practice
- Linear Algebra

(2025 Winter)

(2024 Winter)

(2021, 2023 Winter)

(2023, 2025, 2026 Spring)

(2022 Winter)

(2022 Spring)

INTERNSHIPS

CryptoLab

South Korea

RESEARCH INTERN (PART-TIME)

Sep. 2025 - Feb. 2026

CryptoLab and FHELab

France

RESEARCH INTERN (FULL-TIME)

Jul. 2025 - Sep. 2025

T-Brothers

Canada

UNDERGRADUATE INTERN

Jan. 2019 - Mar. 2019

- Developed in-house applications for controlling the flow of stocks.

EDUCATION PROGRAM

University of Southern California (USC)

Los Angeles, USA

SW PIONEER PROGRAM

Jul. 2019 - Aug. 2019

Education

Integrated Ph.D. in Mathematical Sciences

Seoul, Republic of Korea

SEOUL NATIONAL UNIVERSITY

Sep. 2021 - Present

- Advisor: Prof. Jung Hee Cheon (from Jan. 2023).
- Focus: Homomorphic Encryption, Privacy-Preserving Machine Learning.

B.S. in Mathematical Sciences and Industrial Mathmatics

Busan, Republic of Korea

PUSAN NATIONAL UNIVERSITY

Mar. 2015 - Aug. 2021

- Summa Cum Laude